Trade Dynamics of Environmental Goods within the Global Energy Economy and Their Impacts on Green Technological Innovation

A complex network analysis

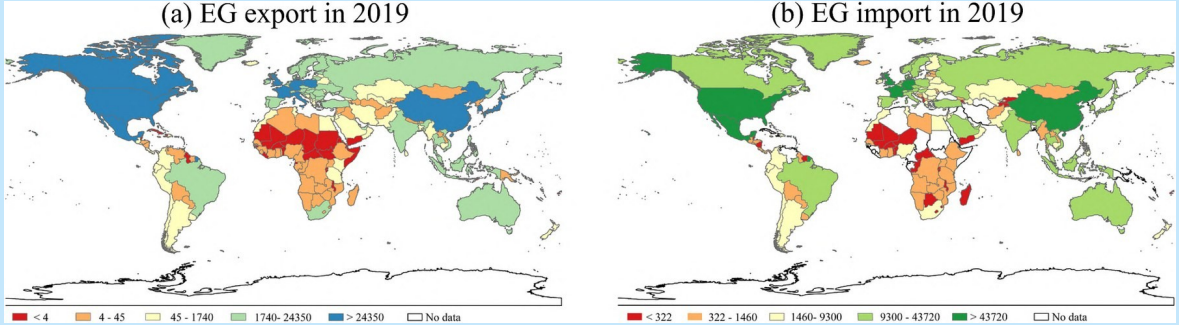
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Prepared for Monash University · Lecturer / Senior Lecturer (EJM 12795)

Trade volume alone misses a country's network advantage



Paper Fig. 2: exports and imports of environmental goods, 2019 (UN Comtrade)

- Exporting environmental goods (EGs) brings learning from demanding foreign markets and scale economies.
- Importing EGs brings technology spillovers, but also competition that can lower gains from innovation.
- Bridging and central trade positions give access to richer flows of information and partners.

>50%

of environmental-goods trade in recent decades is renewable energy plant and wastewater management equipment (OECD CLEG)

OECD and BRICS lead EG trade; Africa engages far less.

Does a country's position in the green-goods trade network shape its green innovation?

Gap: trade-network studies cover fossil fuels, renewables, critical minerals and recycling, but rarely environmental goods. Empirical EG studies measure trade volume, which does not reveal intermediary or central roles.

HYPOTHESES

H1 Stronger trade strength and better network positions within the EGTN promote national green innovation.
H2 Stricter environmental regulations are beneficial to green innovation.

CONTRIBUTION

1. One of few studies to map and visualise the structure and evolution of the global EG trade network.
2. Links network position and environmental regulation to green innovation, using panel quantile regression to uncover non-linear effects.
3. Targeted policy insights for countries at different stages of green-technology development.

Data: a global trade network and a 32-country panel

TRADE NETWORK (EGTN), 2002–2021

- Directed, weighted; countries are nodes, trade value is edge weight
- ~234 countries/regions, 11 EG types (OECD CLEG, HS 2007), UN Comtrade imports

GREEN INNOVATION

- Share of world environment-related inventions (OECD Green Growth)
- Robustness: inventions per million residents, patent family size of at least 2

POLICY AND CONTROLS

- EPS index 0–6 (OECD), split into market and non-market sub-indices
- GDP per capita, population, industry, urbanisation, fixed capital (WDI)

ESTIMATION PANEL, 2002–2019

- 27 OECD + 5 BRICS, balanced, 2002–2019
- 576 observations (486 OECD, 90 BRICS); limited by EPS availability

Three-step empirical strategy

01

NETWORK TOPOLOGY

- Degree, strength and degree distribution
- Density, average path length, clustering
- Betweenness = intermediary; closeness = centre; eigenvector = influence

02

PANEL QUANTILE REGRESSION

- QRPD with non-additive fixed effects (Powell, 2022)
- Quantiles 20, 40, 60, 80 of green innovation
- Controls plus year and country fixed effects

03

ENDOGENEITY AND ROBUSTNESS

- Shift-share IV and non-market EPS as instruments
- Two-step system GMM with lagged innovation
- Alternative proxy; EPS split; OECD vs BRICS

$$\ln AI_{it} = \beta_0 + \beta_1 \ln Network_{it} + \beta_2 EPS_{it} + \Sigma \beta_j X_{jit} + \sigma_i + \sigma_t + \varepsilon_{it}$$

The network became denser and more tightly connected

1.56 → 1.35

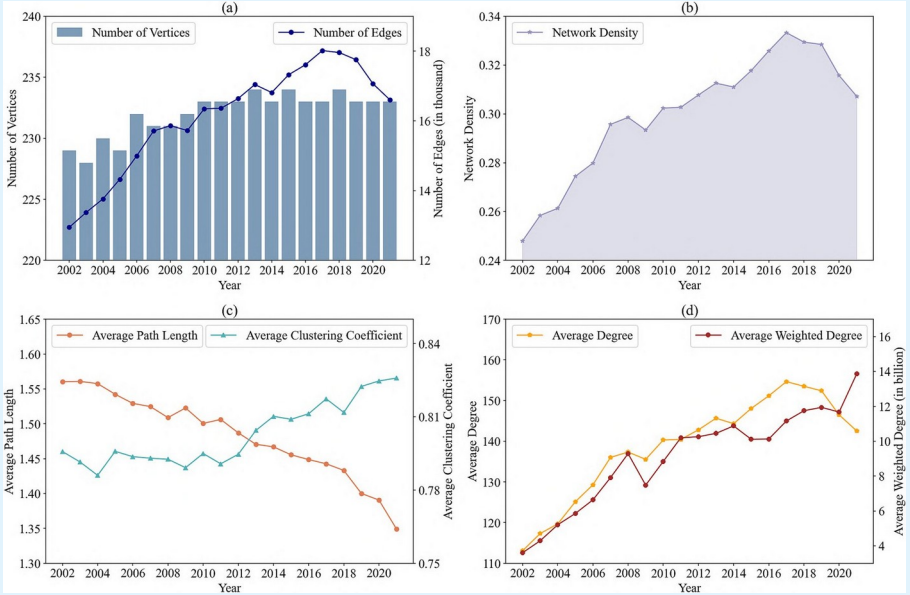
average path length, 2002 to 2021

113 → 155

average trade partners, 2002 to 2017 (143 in 2021)

0.80 → 0.83

average clustering coefficient, 2002 to 2021



Paper Fig. 6: temporal changes of the EGTN, 2002–2021

Density rose until 2018, then fell back to 0.307 by 2021.

A few hubs bridge and steer the network

OECD

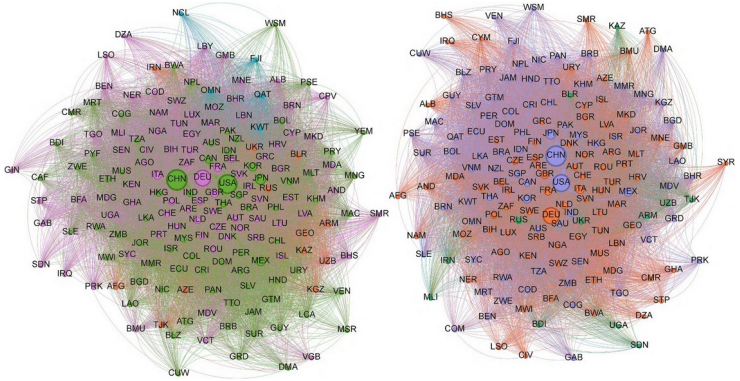
France, Germany, the US and the UK are bridges, central and influential; long-standing top traders include the US, Germany and Japan.

EMERGING

South Africa, China, India and Mexico have served as important bridges since 2006; South Africa ranks high on closeness in all years.

CHINA

Leads trade-strength rankings from about 2010, but its influence and information reach in the network remain limited.



Paper Fig. 5: directed EGTN, 2019 (left) and 2021 (right); node size = trade strength

Few nodes have very high degree: a hub structure, but weaker than a perfect scale-free network.

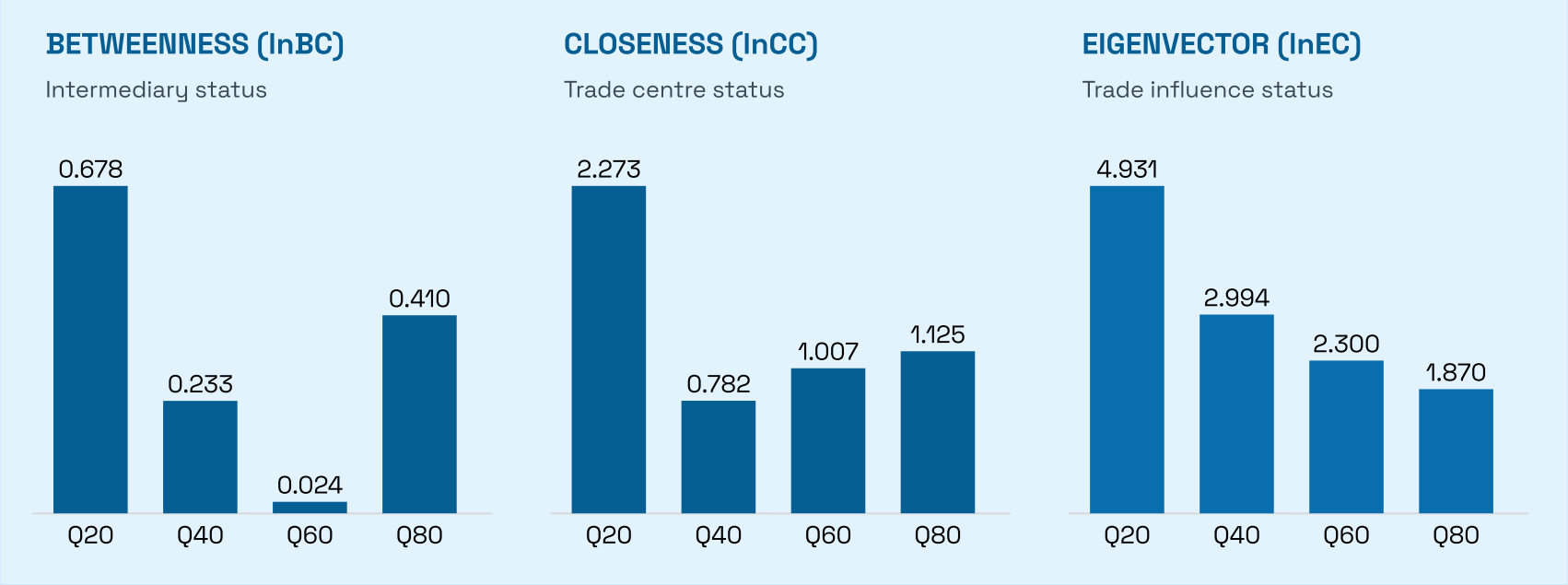
Trade-volume effects change sign across quantiles; position effects stay positive

QRPD coefficient	Q20	Q40	Q60	Q80
lnstrengthin import strength	0.172***	0.228***	-0.183***	-0.092***
lnstrengthout export strength	0.563***	0.368***	0.129***	0.079***
lnstrength total strength	0.499***	0.305***	0.026	-0.108***
lnBC betweenness	**0.678*** **	**0.233*** **	**0.024***	**0.410*** **
lnCC closeness	**2.273*** **	**0.782*** **	**1.007*** **	**1.125*** **
lnEC eigenvector	**4.931*** **	**2.994*** **	**2.300*** **	**1.870*** **

Paper Table 5. Pooled sample, 576 obs.; each model adds EPS, controls, two-way FE. *** p<0.01, ** p<0.05, * p<0.1

- Imports help weaker innovators (spillovers) but hurt leaders (competition).
- Export gains are positive everywhere but shrink at higher quantiles.
- All three centrality measures are positive at every quantile.
- OLS with fixed effects found only export strength significant.

Bridges and centre: U-shaped gains. Influence: gains fade among leaders



QRPD coefficients from paper Table 5 (Fig. 10); all significant at 1%, except lnBC at Q60 (10%).

Position effects survive endogeneity and measurement checks

SHIFT-SHARE IV

6 of 6

network coefficients positive and significant ($p < 0.05$). Hansen J not rejected; $F > 10$ except lnEC (5.95). E.g. lnCC 5.035**.

SYSTEM GMM

3.626**

lnEC; lnBC = 0.375**. Lagged innovation is positive and significant: path dependence. Trade-intensity terms not significant.

ALTERNATIVE PROXY

lnCAP

Inventions per capita: export strength and all centralities positive at all four quantiles; U-shapes for lnBC and lnCC hold.

Paper Tables 6–8 and Fig. 11; N = 544 in IV and GMM models.

Environmental policy helps in OECD; market-based tools may dampen BRICS innovation

Pooled: EPS is positive and significant in OLS and in most quantile models, in line with the Porter Hypothesis.

OECD (27 countries)

- Market and non-market EPS are positive in all six models (Table 9).
- Significant in most models, e.g. market-based EPS 0.056**.
- Mean EPS 2.81, more than twice the BRICS mean of 1.13.

BRICS (5 countries)

- Market-based EPS is negative; significant at 10% in half the models (e.g. −0.415*).
- Non-market EPS is negative but insignificant in all models.
- Possible channel: political costs of market-based instruments.

Policy implications and next steps

FOR POLICY

- Pursue trade agreements that cut tariff and non-tariff barriers on critical environmental goods.
- Target the quality of trade ties, not only volume: trade more with influential, innovative partners.
- Intermediary countries can bridge core and periphery through joint projects and research partnerships.
- BRICS: public–private–academic collaboration to offset adverse effects of market-based tools.

LIMITATIONS AND FUTURE WORK

- Patent-based proxy misses non-patentable or proprietary innovation.
- Sample of 32 countries set by EPS data; extend to more nations.
- Build industry-level indicators; add environmental R&D, energy policy and green finance.

Conclusion

01 **Position matters, not only volume**

Intermediary, central and influential positions in the EG trade network raise green innovation at every quantile.

02 **Effects are non-linear**

U-shaped for betweenness and closeness; the influence (eigenvector) effect is largest for weaker innovators and fades among leaders.

03 **Policy design depends on context**

Stricter EPS helps overall and in OECD; market-based instruments may dampen innovation in BRICS.

THANK YOU

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Yu, Y., Qayyum, M., & Li, S. (2024). Trade dynamics of environmental goods within global energy economy and their impacts on green technological innovation: A complex network analysis. Energy Economics, 140, 107957. doi:10.1016/j.eneco.2024.107957